DATA CENTER SIMULATION PERFORMANCE REPORT

Workload Profile: NORMAL_DAY | Timestamp: 2026-08-23 18:45:51

OVERALL SLA VERDICT: FAIL (3/4 Checks Satisfied)

1. Experiment Summary & Identification

Parameter	Value
Profile / Workload	normal_day
Execution Duration	600.0 s
Start Time (T1)	1787490349.714588
End Time (T2)	1787490949.714588
Total Requests Executed	41
Primary Bottleneck Candidate	Application / Endpoint Latency
Statistical Confidence	LOW_SAMPLE_SIZE
Failure Classification	SLA_NON_COMPLIANCE

2. SLA Scorecard & Compliance Targets

SLA Metric	Measured Value	Target Limit	Result
P95 Response Time	1900.0 ms	500.0 ms	FAIL
Error Rate	0.0 %	1.0 %	PASS
Peak CPU Utilization	4.72 %	80.0 %	PASS
Peak Memory Utilization	18.37 %	90.0 %	PASS

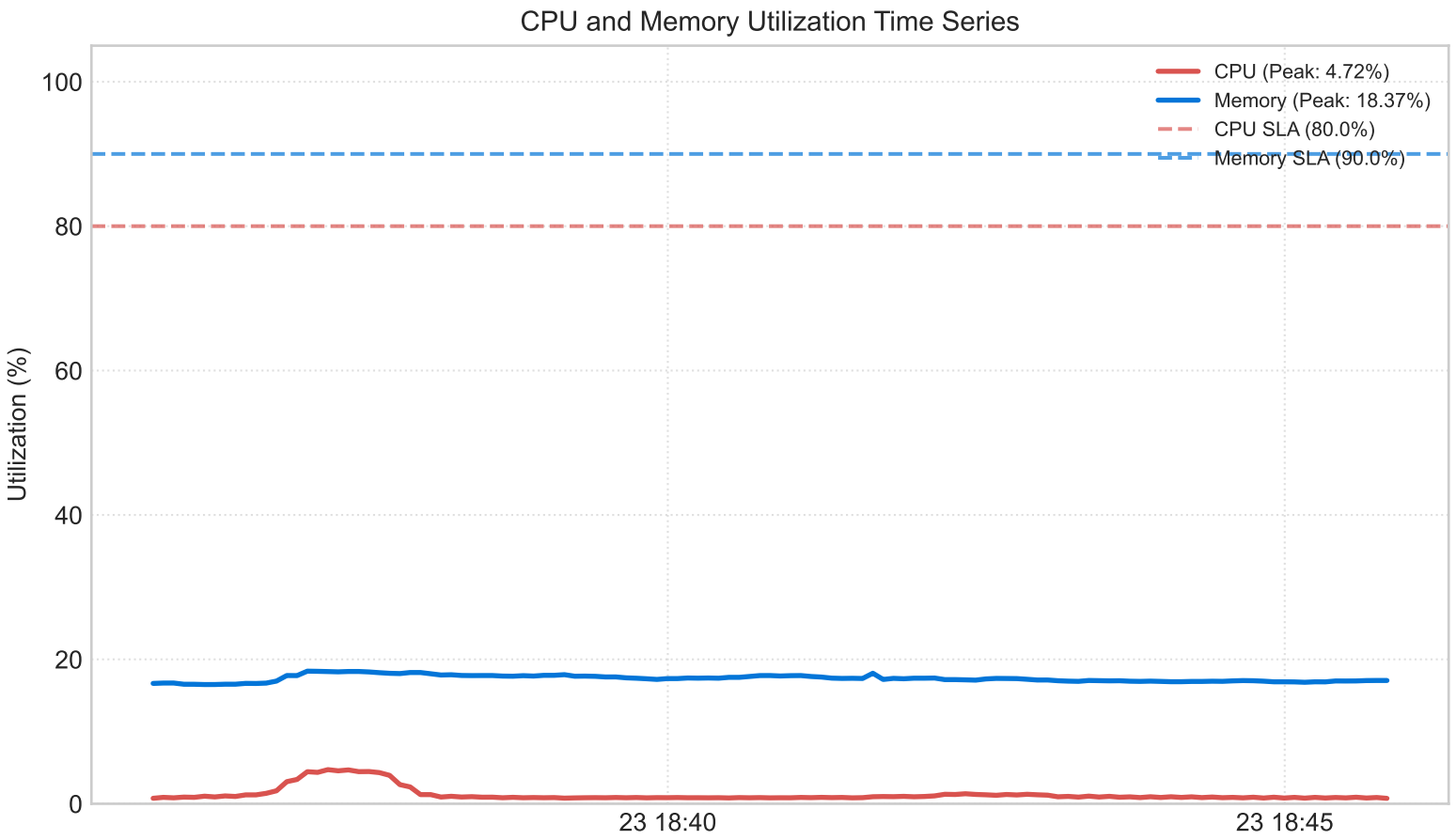
3. Executive Conclusion & Synthesis

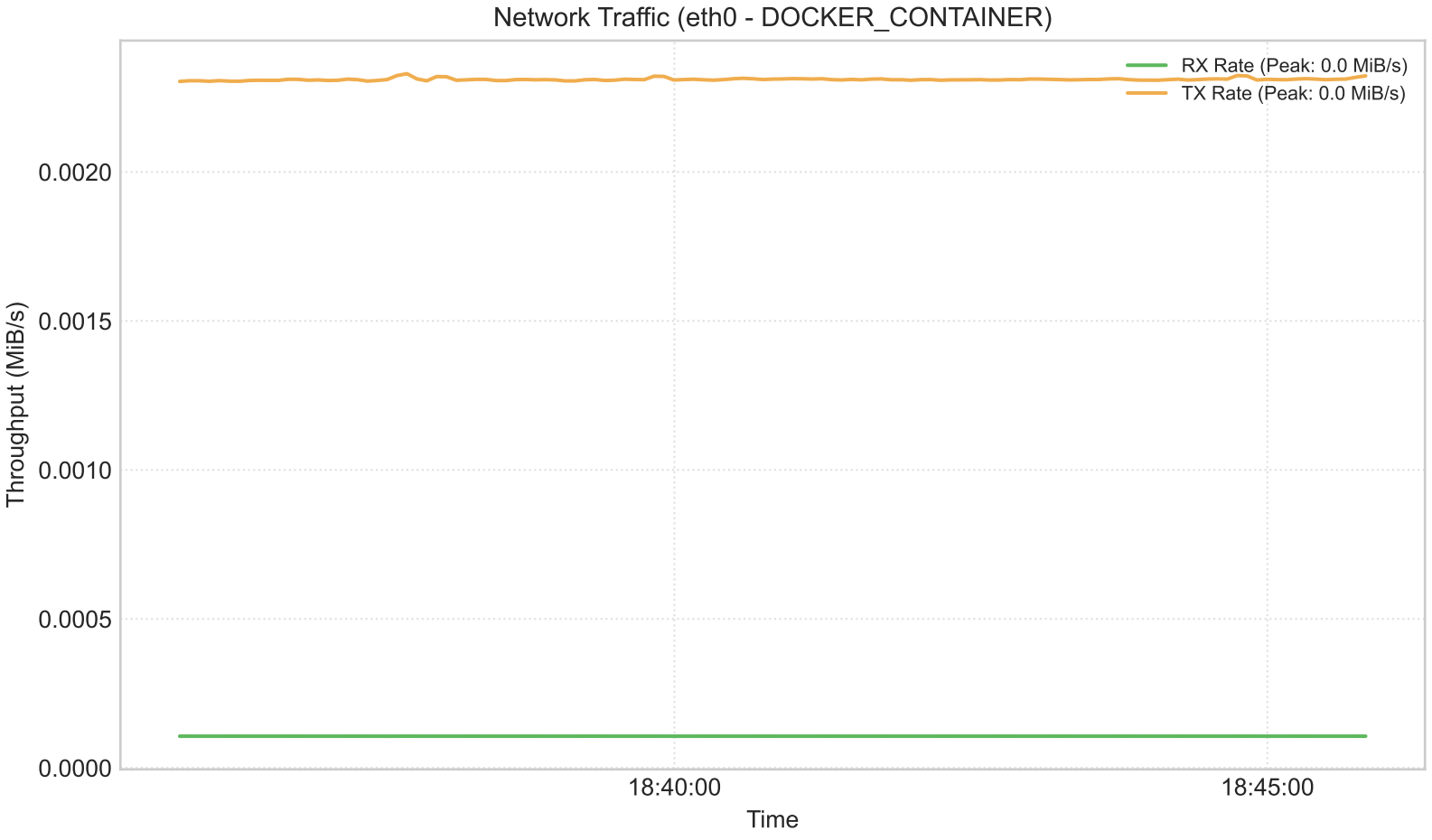
The experiment executed 41 requests with overall SLA status 'FAIL' (3/4 checks satisfied). No container crashes or service interruptions were detected by the watchdog. Primary observed degradation candidate identified as 'Application / Endpoint Latency' (tentative due to low request volume).

4. Diagnostic Observations & Failure Evidence

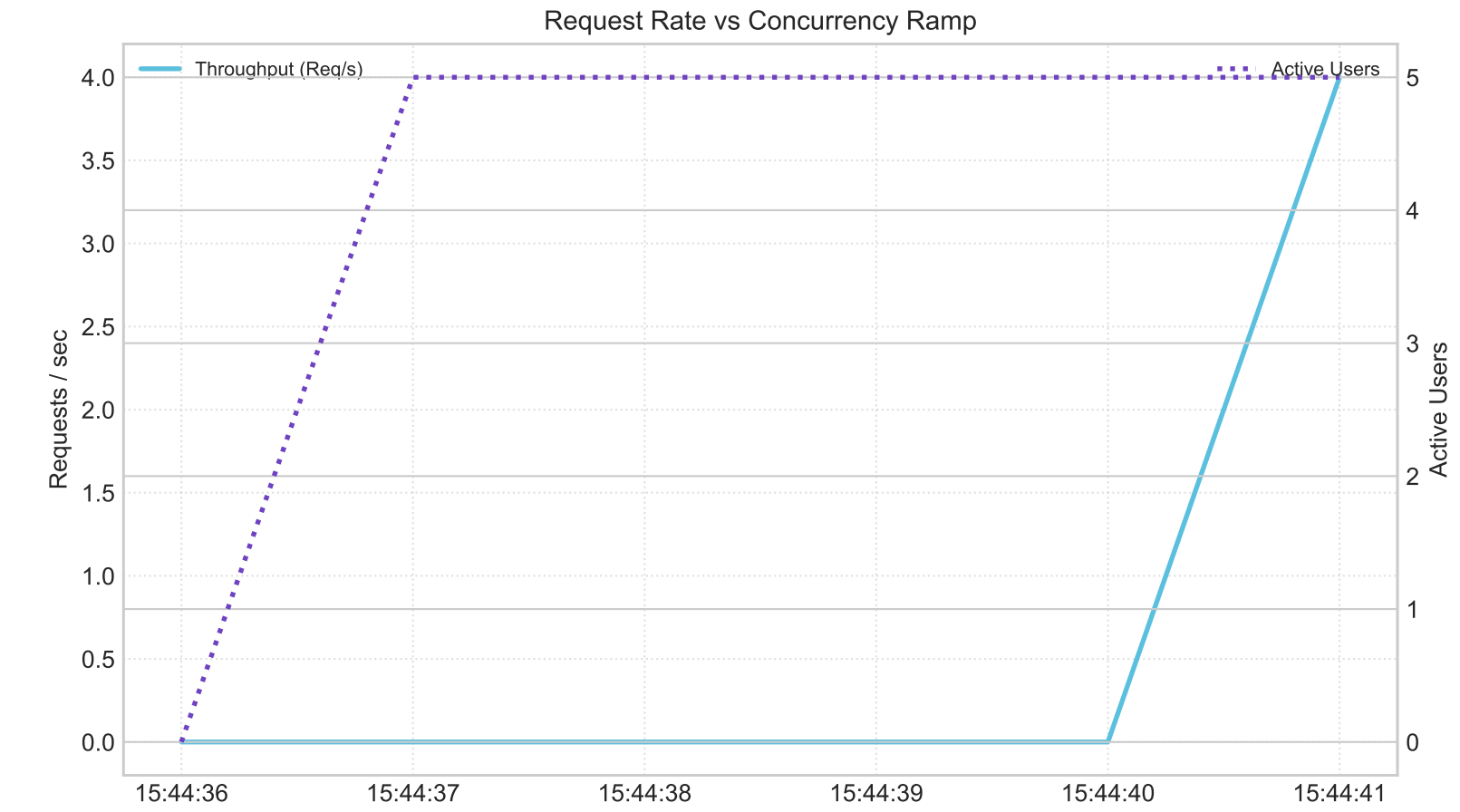
- Low request volume (41 requests); statistical latency metrics are tentative.
- P95 response time (1900.0 ms) violated SLA target (500.0 ms).
- Endpoint bottleneck identified on 'POST /login' with P95 of 1900.0 ms.
- SLA Target Violated: P95 Response Time: measured 1900.0 ms exceeds limit 500.0 ms

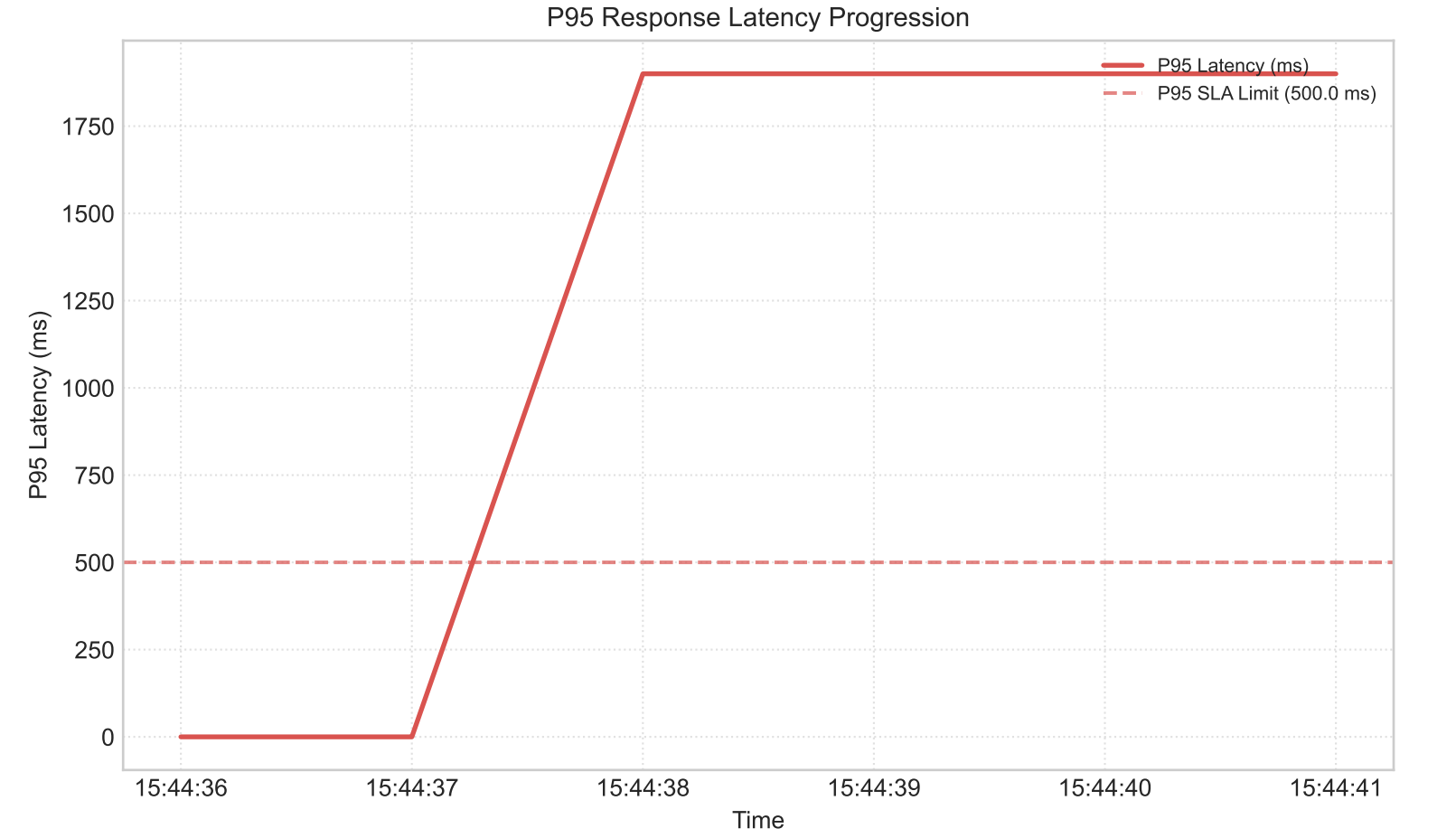
System & Hardware Resource Dynamics





Locust Workload & Latency Profile





Endpoint Breakdown & Chronological Event Timeline

1. HTTP Endpoint Performance Breakdown

Endpoint	Requests	Failures	Avg Latency	P95 Latency	RPS
GET /api/data	14	0	76.34 ms	140.0 ms	1.56
GET /health	9	0	10.55 ms	19.0 ms	1.0
POST /login	5	0	1885.21 ms	1900.0 ms	0.56
GET /me	12	0	13.19 ms	28.0 ms	1.34
POST /profile/update	1	0	61.69 ms	62.0 ms	0.11

2. Chronological Event Timeline

Time	Event Source & Type	Category	Severity	Description
15:44:36	[LOCUST] LOAD_STARTED	OBSERVED	INFO	Workload generation began with 0 users
15:44:37	[LOCUST] LOCUST_PEAK_USERS	OBSERVED	INFO	Locust concurrency reached peak of 5 active users
15:44:38	[SLA_ENGINE] SLA_BREACH_DETECTED	INFERRED	CRITICAL	SLA Target Breached: P95 Response Time was 1900.0 ms (L...
15:44:41	[LOCUST] LOCUST_PEAK_RPS	OBSERVED	INFO	Locust throughput reached peak of 4.0 req/s
18:35:49	[RUNNER] EXPERIMENT_START	OBSERVED	INFO	Experiment started at T1 (1787490349.714588)
[PROMETHEUS]	18:37:04 PROMETHEUS_MEMORY_PEAK	OBSERVED	INFO	Memory utilization peaked at 18.37%
[PROMETHEUS]	18:37:04 PROMETHEUS_CPU_PEAK	OBSERVED	INFO	CPU utilization peaked at 4.72%
[PROMETHEUS]	18:37:04 NETWORK_PEAK_RX	OBSERVED	INFO	Network receive rate peaked at 0.0 MiB/s on eth0
[PROMETHEUS]	18:37:04 NETWORK_PEAK_TX	OBSERVED	INFO	Network transmit rate peaked at 0.0 MiB/s on eth0
18:45:49	[RUNNER] EXPERIMENT_END	OBSERVED	INFO	Experiment completed at T2 (1787490949.714588)